

Evaluating Convolutional Neural Networks for Earthquake-Induced Landslide Susceptibility Mapping: A Case Study of the 2015 Gorkha Earthquake Using Pre-Event Satellite Imagery

INTRODUCTION

Can convolutional neural networks (CNNs) effectively predict coseismic landslide susceptibility in regions triggered by the 2015 Gorkha earthquake using pre-earthquake satellite imagery?

The 2015 7.8 Gorkha earthquake caused over 25,000 landslides across central Nepal, as mapped by Roback et al. (2018) using high-resolution satellite imagery and GIS techniques. Their study provided a comprehensive polygon-based inventory distinguishing landslide source and deposit areas, revealing that the landslide distribution was influenced by the confluence of steep slopes, high seismic energy release, and annual precipitation. Previous work on landslide susceptibility mapping has shown the utility of machine learning (ML) models like random forests, logistic regression, and support vector machines (Bui et al., 2020 & Youssef et al., 2020), but CNNs have primarily been applied to landslide detection rather than susceptibility mapping. For instance, Ghorbanzadeh et al. (2019) utilized CNNs to identify landslide-affected areas with high accuracy but did not incorporate susceptibility modeling. This highlights a key knowledge gap: whether CNNs, trained on pre-earthquake satellite data, can predict coseismic landslide susceptibility maps that align with post-event observations. To address this, I use the aforementioned landslide inventory data (Roback et al., 2018) as ground truth data and train a CNN model using multi-band pre-earthquake layers (NDVI, DEM, soil moisture) to predict landslide susceptibility and evaluate its performance using Average Precision, ROC-AUC, and pixel-wise F1 scores. This approach tests the feasibility of CNN-based susceptibility mapping for earthquake-induced landslides.

STUDY SITE

The study area in central Nepal (Figure 1) provides an ideal location for testing deep learning approaches to landslide susceptibility mapping due to the availability of a comprehensive, freely accessible earthquake induced landslide inventory compiled by Roback et al. following the 2015 Gorkha earthquake. The site's diverse topographic and geological conditions within the Himalayan Mountains present complex relationships between landslide triggering factors, as documented in Roback et al.'s work. While these conditions create challenging terrain for analysis, the availability of comprehensive landslide inventory data and satellite imagery makes this an ideal test case for evaluating whether CNN models can capture nonlinear relationships that simpler models may struggle with. Additionally, Google Earth Engine provides free access to the necessary pre-earthquake satellite imagery required for this analysis.

METHODS

1. Data Collection and Preprocessing

1.1 Ground Truth Landslide Mapping

The ground truth landslide inventory, mapped by Roback et al. (2018), was formatted as polygon shapefiles. Using ArcGIS Pro, Polygons were simplified and corrected to ensure geometric integrity before converting to a binary raster mask, where pixel values represent: 1 = landslide presence, 0 = no landslide. The landslide raster map was then resampled at 30x30m resolution to match satellite data granularity which ensured consistency across all input channels, although some spatial quality was lost.

1.2 Remote Sensing Data Acquisition

Pre-earthquake satellite data was collected through Google Earth Engine, focusing on the following parameters:

a. Normalized Difference Vegetation Index (NDVI)

NDVI quantifies vegetation density values ranging from -1 to +1, where higher values indicate denser vegetation. NDVI was calculated using the following equation:

$$NDVI = (NIR - RED) / (NIR + RED)$$

Where NIR = Near-infrared reflectance and RED = Red band reflectance, both of which were extracted from LANDSAT data.

b. Digital Elevation Model (DEM)

DEM represents terrain elevation above sea level in meters, from which slope (in degrees) and aspect (in degrees from north) were derived using ArcGIS Pro's surface analysis tools.

Low scale datasets (soil moisture and land cover) were excluded due to resolution constraints. All layers were standardized to 30x30m resolution and converted to raster format.

1.3 Image Processing

The processed data was divided into 256x256 pixel tiles with each tile containing multiple layers (Figure 2) and split into training, validation and testing batches. The final dataset consisted of: 47 Training images, 9 Validation images and 10 Test images. To address the limited dataset size, geometric augmentation was randomly applied during training. Missing values (NaN) were mapped to 0.

2.1 Convolutional Neural Network Design

The input to the model was the aforementioned tiles of shape 256x256 image tile, where the 4 channels correspond to: NDVI, DEM, Slope, and Aspect (Figure 2). The model comprises downsampling blocks for feature extraction and upsampling blocks for reconstruction:

- Encoder: Five convolutional layers with ReLU activations, Batch Normalization, and MaxPooling to reduce spatial dimensions.
- Decoder: Five transposed convolution layers (upsampling) with ReLU activations to restore the spatial resolution and predict per-pixel probabilities.

The final output layer applies a Sigmoid activation to produce a probability map of landslide susceptibility.

2.2 Loss Function

The CNN model was trained using the pytorch's weighted binary cross-entropy with logits loss function:

$$L = - (1/N) \sum_{i=1}^N [w_1 y_i \log(p_i) + w_0 (1 - y_i) \log(1 - p_i)]$$

Where: y_i = ground truth label (0 or 1), p_i = predicted probability, w_1 = positive class weight (set to 60,000 for positive class to address severe class imbalance), $w_0 = 1$.

3. Model Evaluation Metrics

The performance of the model was assessed using a combination of widely accepted metrics to evaluate classification accuracy, precision, and reliability. These include Precision, Recall, the F1 Score, Average Precision (AP), and the ROC-AUC (Receiver Operating Characteristic - Area Under Curve) score.

3.1 Performance Metrics

Precision measures the proportion of correctly predicted positive pixels (true positives) among all predicted positive pixels. It is defined as:

$$(1) \text{ Precision} = TP / (TP + FP)$$

where TP represents true positives, and FP represents false positives.

Recall, also referred to as the True Positive Rate (TPR) or sensitivity, calculates the proportion of actual positive pixels correctly identified by the model:

$$(2) \text{ Recall} = TP / (TP + FN)$$

where FN corresponds to false negatives.

To balance Precision and Recall into a single performance measure, the F1 Score is used. The F1 score is the harmonic mean of Precision and Recall, ensuring that both contribute equally:

$$(3) F1 = 2 \cdot (Precision \cdot Recall / (Precision + Recall))$$

In addition, Average Precision (AP) summarizes the Precision-Recall curve by calculating a weighted mean of precision values at various recall thresholds. It is given by:

$$(4) AP = \sum_n (R_n - R_{n-1}) P_n$$

where R_n is the recall at the n -th threshold, and P_n is the corresponding precision.

The ROC-AUC score evaluates the model's ability to discriminate between positive and negative classes across different classification thresholds. It is computed as the area under the curve plotting Recall (Equation 2) against the False Positive Rate (FPR), defined as:

$$(5) FPR = FP / (FP + TN)$$

where TN represents true negatives. The ROC-AUC score provides a threshold-independent measure of the model's overall classification performance.

3.2 Optimal Threshold Selection

While the ROC-AUC score evaluates model performance across all thresholds, a single optimal classification threshold must be chosen for practical application. In this study, the threshold was selected by maximizing the F1 score (Equation 3) across all possible thresholds. Specifically, the process involved the following steps:

1. For each threshold t , Precision and Recall were computed based on Equations (1) and (2).
2. The corresponding F1 score at threshold t was calculated using Equation (3).
3. The threshold t^* that produced the maximum F1 score was identified as the optimal

Results

The training loss, graphed in figure 4, decreased from an initial value of 3000 to 900 within the first 5 epochs, followed by a continued decrease until epoch 15 where it reached approximately 500. Between epochs 15 and 30, the training loss maintained a steady state between 450-500. The validation loss, graphed in figure 5, was only calculated every 5 epochs and started at 800 and gradually decreased to 700 by epoch 10. From epochs 10 to 30, the validation loss remained between 680-750, with minor fluctuations throughout this period.

The test set example in Figure 3 displays three components: the input satellite imagery, ground truth landslide locations, and the model's predictions. The ground truth map shows landslide occurrences concentrated in the eastern portion of the image, appearing as dark red patches. The model's prediction heat map produces larger blocks of varying intensity, with the highest intensity predictions (darkest red) corresponding to areas in the eastern section where ground truth landslides are present.

The model's performance metrics are summarized in Table 1. The ROC-AUC score reached 0.919. At the optimal threshold of 0.5022, the model achieved a precision of 0.0198 and recall of 0.4080, resulting in an F1 score of 0.0378. The average precision across all thresholds was 0.0161. These metrics were calculated on the test set using the evaluation protocol described in the methods section.

Discussion

The CNN model achieved a high AUC score of 0.919, indicating it can effectively rank landslide probability, but the extremely low precision (0.0198) and mediocre recall (0.4080) reveal that the model fails to make reliable binary predictions about landslide occurrence and vastly over predicts landslides. This AUC performance aligns with several previous machine learning approaches for landslide susceptibility mapping. Tien Bui et al. (2020) reported AUC values of around 0.95 using a different Deep Learning Neural Network model on a study area in Vietnam, while Youssef et al. (2020) achieved AUC scores of 0.89 to 0.95.1 using different non deep learning ML models such as random forest and SVM in Saudi Arabia. However, our precision metrics are notably lower than those reported in comparable studies - for instance, compared to Bui et al. (2020), my model significantly underperforms in terms of precision (0.0198 vs. 85.56% or higher) and recall (40.8% vs. 82.0–95.02%), highlighting challenges in accurately identifying and minimizing false positives and negatives at the chosen threshold.

Our results particularly diverge from prior work in terms of precision and recall balance likely stemming from the following factors:

1. Extremely skewed class distribution required a high weight factor of 60,000
2. I was limited to 30m resolution data, while studies reporting higher precision like Tien Bui et al. (2016) utilized 10m resolution inputs
3. Computational constraints limited hyperparameter optimization

However, even with more exploration into different CNN model structures and hyperparameter selection, the stagnation of training loss around 500 indicates to me that the more likely limiting factor is the small training dataset size and lack of resolution. Furthermore, the model is likely not very robust or generalizable to other locations, given its small training data set size.

Conclusion

Deep learning approaches for landslide susceptibility mapping remain an important area of research, with this study demonstrating both the potential and current limitations of CNN-based approaches. While our CNN model achieved a strong discriminative ability with a ROC-AUC score of 0.919, the low precision (0.0198) and recall (0.4080) highlight significant challenges in practical application, potentially due to data limitations. As higher resolution satellite data becomes increasingly available and more comprehensive landslide inventories are developed, particularly following major triggering events like earthquakes, CNN-based approaches may become more viable for susceptibility mapping. Future work should focus on building larger, higher-quality training datasets to better capture the complex relationships between pre-event conditions and landslide occurrence. Additionally, tracking the timing and triggers of landslides will be crucial for advancing both deep learning and traditional approaches to susceptibility mapping.

APPENDIX

Fig. 1 The study area is located in central Nepal, shown by the red rectangular overlay which encompasses Kathmandu and extends into the Himalayan mountain range. This region corresponds to the maximum extent of coseismic landslides mapped by Roback et al. following the 2015 Gorkha earthquake. All remote sensing analyses, including NDVI calculations and DEM-derived products, were confined to this rectangular extent to align with the available landslide inventory data. (Base imagery obtained from Google Earth Engine's satellite collection.)



Figure 2. Input data layers used for landslide prediction: (A) NDVI showing vegetation coverage, with values ranging from -1 (no vegetation) to +1 (dense vegetation), (B) DEM representing elevation in meters above sea level, (C) slope map derived from DEM showing terrain steepness in degrees, and (D) aspect map indicating slope orientation in degrees from north. All layers were processed to 30x30m resolution and combined as input channels for the CNN model. This is the 5th input image in the test set.

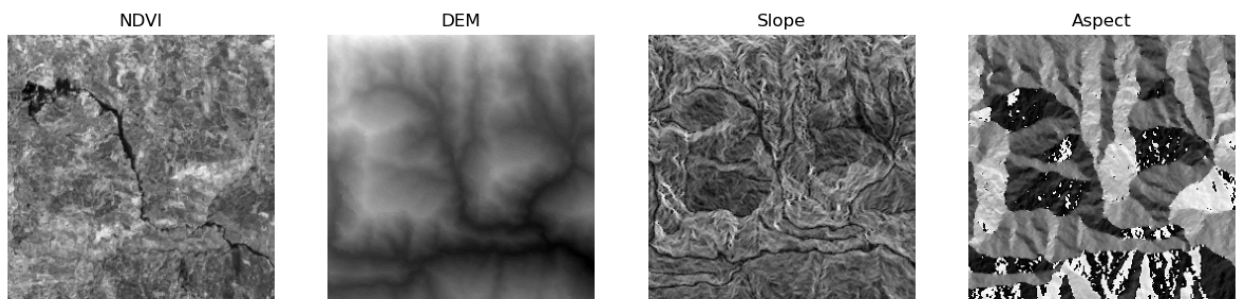


Fig. 3 Example prediction from the test dataset showing (left) the input satellite imagery with only the NDVI channel displayed in green, (middle) the ground truth landslide map with landslides in dark red, and (right) the model's prediction mapped to a heat map where darker red indicates higher predicted landslide probability.

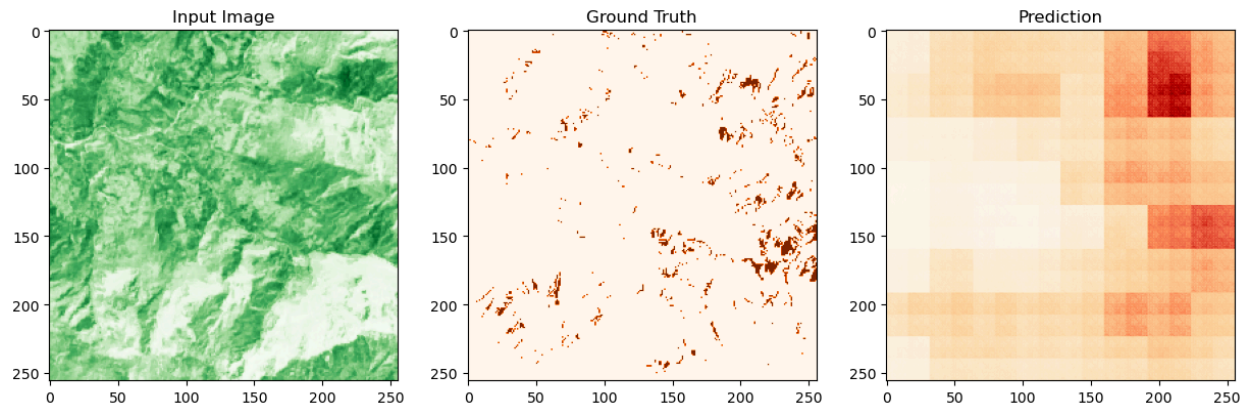


Fig. 4 Training loss curve over 30 epochs showing rapid initial convergence in the first 5 epochs followed by gradual stabilization, with the final loss stabilizing around 500.

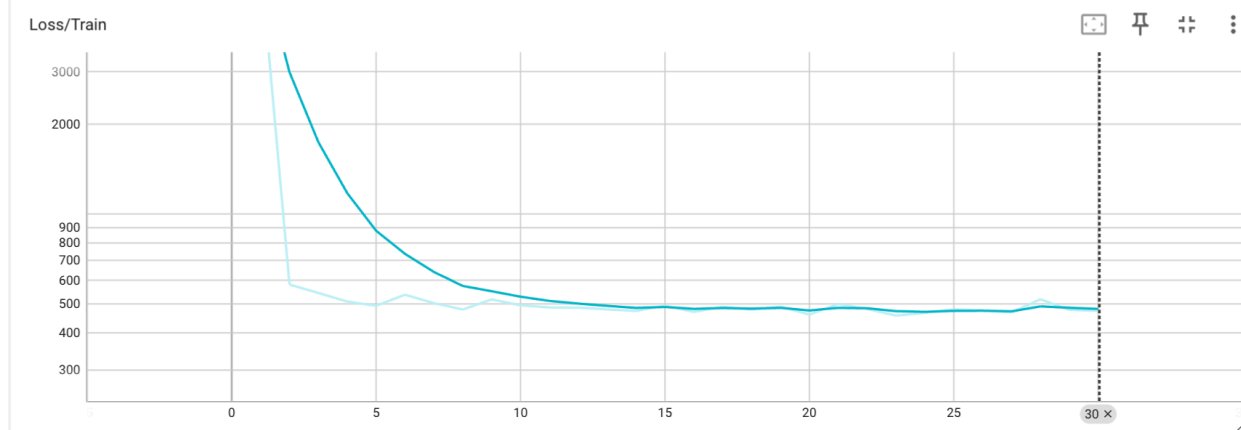


Fig. 5 Validation loss curve calculated every 5 epochs over 30 epochs demonstrating stable model performance with minimal fluctuation, maintaining a loss value around 700 throughout most of the training process.

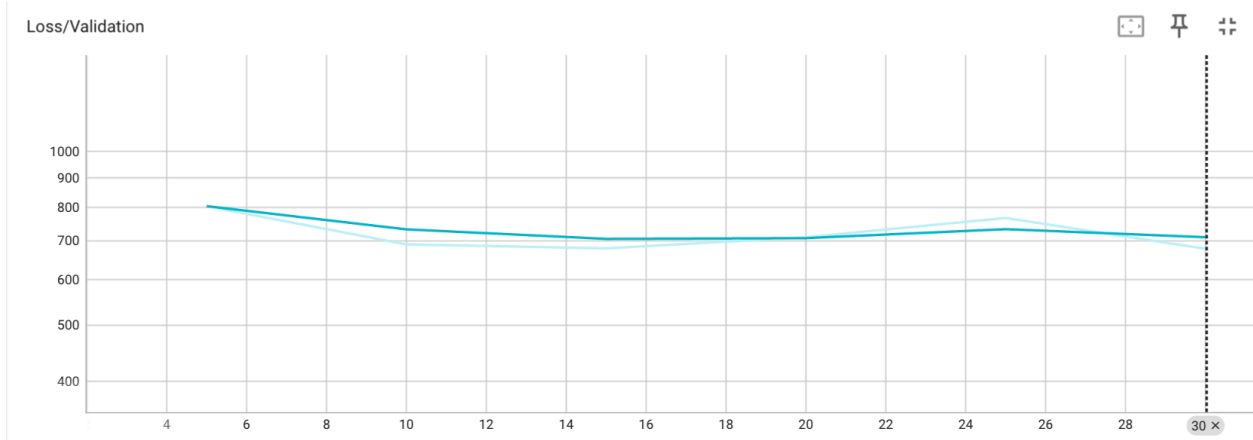


Table 1. Summary of model evaluation metrics evaluated on the test set showing strong discriminative ability (ROC-AUC: 0.919) but limited precision (0.0198) due to class imbalance, with performance metrics calculated at the optimal operating threshold of 0.5022.

Average Precision	0.0161
Optimal Threshold	0.5022
Precision at optimal threshold	0.0198
Recall at optimal threshold	0.4080
F1 Score at optimal threshold	0.0378
ROC-AUC	0.9190

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